

this technology to run on any device while bypassing the restrictions of browser-only application functionality. Cross-platform software basically compiles a single application source code into the native code that runs on different OSs, letting developers work on the same source code.

Requirements for mobile app development

Let's have a look at all that a mobile app developer needs to take care of before starting off.

1. Understand and locate the problem or pain points that people experience and come up with the simplest, most effective and unique solution for this. Some market research may be needed for this.
2. Set targets and build a strategic plan to work upon according to the requirements of the client. This helps to avoid any unexpected changes in between and complete the development task in time.
3. Choose the right technology as this really plays a decisive role in the long run. Technology is really evolving at a rapid pace and developers always want to opt for the latest technology during app development. They need to be in sync with the latest technology trends if they want to make a strong impression in the application store. It's always recommended that a mobile app should first be focused at one specific platform. It can be launched on multiple platforms, subsequently.
4. Always go for user-centric designs as the user experience (UX) is an extremely crucial consideration in the design of any mobile application. The UX is closely linked to other aspects like the user input mechanism and user interaction design.
5. Prioritise the efficiency and performance of the mobile app that needs to be developed. Rapid response time and a stable performance without crashing are the key performance criteria by which any user ranks a mobile application.
6. Define the target audience or the end users of the mobile application as they will impact how you develop your mobile app. All questions, such as the end users of the app, and how it's going to help them, need to be answered well before hand. If we meet our users' expectations, our smartphone application is likely to get more popular.
7. Plan for an effective testing process to be followed before the app actually hits the market. Keep in mind the performance, efficiency and load testing of the mobile app.
8. Help people save time and money, as that increases the chance of your app being downloaded. If you are successful in this, then you have nailed it, and your application is likely to get the deserved attention.
9. Know your budget in advance. Consider all the steps included in the development process which involve money, and properly allocate a budget for each stage.

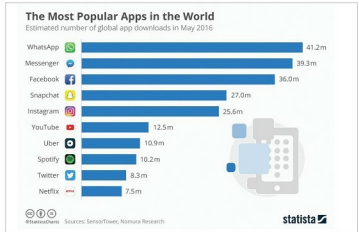


Figure 3: The most popular mobile applications (Information source: Statista)

Open source programming languages used for mobile application development

We know that a mobile app is nothing but a piece of code which performs the specified task, and code can be written in multiple programming languages. Now let's look at some of the open source programming languages widely used for mobile app development.

Objective-C: This is the primary open source programming language used for different iOS applications. Objective-C was actually chosen by Apple to build apps that are scalable and robust. In spite of being a C-language superset, it does actually have a number of functions that precisely deal with I/O, graphics and display functions. Objective-C is completely integrated into all the iOS and MacOS frameworks. However, it's now being slowly replaced by a more powerful open source language called Swift.

HTML5: HTML5 is one of the ideal open source programming languages if you are looking to build a Web front-end app for mobile devices. It makes various data types quite simple to insert, rationalises input parameters and accounts for different screen sizes, and also levels the browser playing field. It's currently supported in different ways by various browsers. If we look at HTML5 from a cost-efficiency point of view, it has the advantage of building on the latest version of HTML, which makes the learning process simpler than that for a completely new language.

Swift: Swift is the latest open source programming language to enter the Apple ecosystem. It is mainly used to write code for Apple's latest APIs, Cocoa and even Cocoa Touch. In spite of the fact that this language is written to work along with Objective-C, the Cupertino based company is always encouraging iOS developers to switch to Swift for complete programming. It's designed to eliminate many of the security vulnerabilities that are associated with Objective-C.

Java: Java is one of the most preferred languages for Android app development. It's an object-oriented programming language which was developed at Sun Microsystems. It can be run in two different ways — by using a browser window, or in a virtual machine without